

Advanced VLSI Design - Lab 1

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1 Schematic and Design of a 5-bit R-2R Ladder Digital to Analog Converter

The aim of this lab was to design a 5-bit DAC using n-well resistance values of $10k\Omega$, with the 2R resistor being implemented as two separate $10k\Omega$ resistors in series. In order to find the output resistance of the DAC, all bit inputs are grounded ($b_0 \rightarrow b_4$). The 2R resistance at b_0 becomes parallel with the 2R resistance tied to ground, resulting in $2R || 2R = R$. This resulting resistance is then in series with with R the next ladder resistor, yielding 2R. The new 2R is then in parallel with the b_1 branch, again yielding R. This pattern repeats identically up the entire ladder. Consequently, the Thevenin resistance (R_{th}) is exactly R.

$$R_{out} = R = 10k\Omega$$

To determine the load drive and delay, all DAC inputs except b_4 were grounded. A pulse source (0 to VDD) was applied to b_4 to predict the delay. The delay can also be calculated as follows:

$$t_d = 0.7RC \Rightarrow 0.7(10k)(10pF) = 70ns$$

1.1 Driving a $10k\Omega$ Load (R_L)

Connecting a $10k\Omega$ load resistor to the V_{out} node creates a voltage divider between the DAC's output resistance and the external load.

$$V_{out,load} = V_{DAC} \cdot \frac{R_L}{R_L + R_{out}} \Rightarrow 0.5 \cdot V_{DAC}$$

The result of this is the ideal output voltage will be halved. Ideally an output buffer would be used in place to prevent such voltage drops from occurring.

2 Simulation Results

2.1 Simple Voltage Divider Results

The following simulation results demonstrate the output voltage of a simple voltage divider using the R-2R ladder topology. The voltage divider layout is also provided. For both cases, the LTspice output is identical.

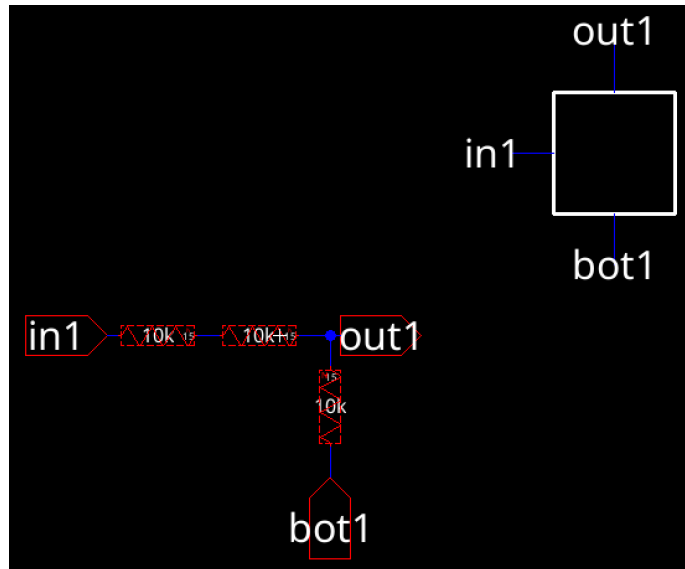


Figure 1: Simple R-2R Voltage Divider Utilizing 10K Resistor and Output Voltage of 1V

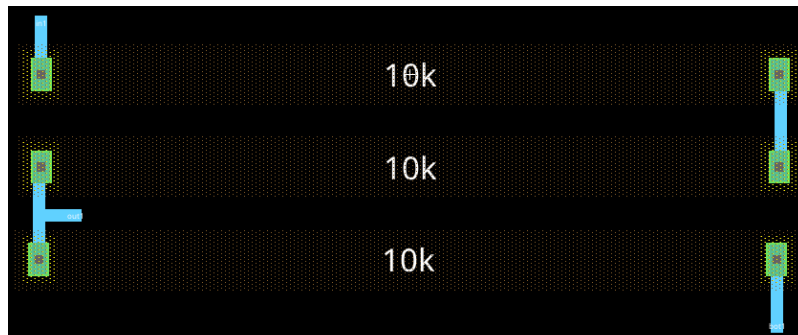


Figure 2: Simple R-2R Voltage Divider Layout

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--- Operating Point ---

V ( v i n ) :           1                v o l t a g e
V ( v o u t ) :         0 . 3 3 3 3 3 3    v o l t a g e
I ( v i n ) :           - 3 . 3 3 3 3 3 e - 0 5    d e v i c e _ c u r r e n t
I x ( x b 0 @ 0 : b o t 1 ) : - 3 . 3 3 3 3 3 e - 0 5    s u b c k t _ c u r r e n t
I x ( x b 0 @ 0 : i n 1 ) :  3 . 3 3 3 3 3 e - 0 5    s u b c k t _ c u r r e n t
I x ( x b 0 @ 0 : o u t 1 ) : 0                s u b c k t _ c u r r e n t

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Figure 3: Voltage Divider Schematic Simulation LTspice Output

2.2 DAC Results

The following images demonstrate the DAC and LTspice results which can be seen on DACsch of the .jelib file. A voltage of 5V was applied to inputs b_0 and b_1 .

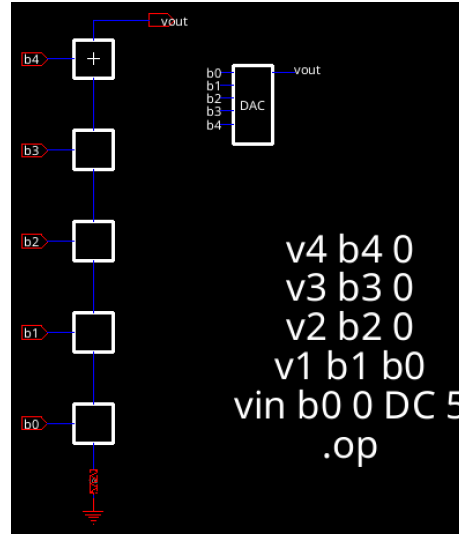


Figure 4: R-2R DAC Schematic

--- Operating Point ---		
V(b0):	5	voltage
V(b1):	5	voltage
V(b2):	0	voltage
V(b3):	0	voltage
V(b4):	0	voltage
V(net#0):	0.703125	voltage
V(net#1):	1.28906	voltage
V(net#11):	1.25488	voltage
V(net#2):	2.51953	voltage
V(net#3):	2.50977	voltage
V(vout):	0.46875	voltage
I(Rresvwell#0):	0.000125488	device_current
I(v1):	-0.000124023	device_current
I(v2):	6.44531e-05	device_current
I(v3):	3.51562e-05	device_current
I(v4):	2.34375e-05	device_current
I(vin):	-0.000248535	device_current
I(xb0#0:bot1):	2.34375e-05	subckt_current
I(xb0#0:in1):	-2.34375e-05	subckt_current
I(xb0#0:out1):	0	subckt_current
I(xb0#1:bot1):	5.85937e-05	subckt_current
I(xb0#1:in1):	-3.51562e-05	subckt_current
I(xb0#1:out1):	-2.34375e-05	subckt_current
I(xb0#2:bot1):	0.000123047	subckt_current
I(xb0#2:in1):	-6.44531e-05	subckt_current
I(xb0#2:out1):	-5.85937e-05	subckt_current
I(xb0#3:bot1):	-9.76563e-07	subckt_current
I(xb0#3:in1):	0.000124023	subckt_current
I(xb0#3:out1):	-0.000123047	subckt_current
I(xb0#4:bot1):	-0.000125488	subckt_current
I(xb0#4:in1):	0.000124512	subckt_current
I(xb0#4:out1):	9.76563e-07	subckt_current

Figure 5: R-2R DAC LTSpice Simulation Results

The following images demonstrate the IC schematic of the R-2R DAC with simplified LTspice code. This schematic can be viewed in DAC_simsch.

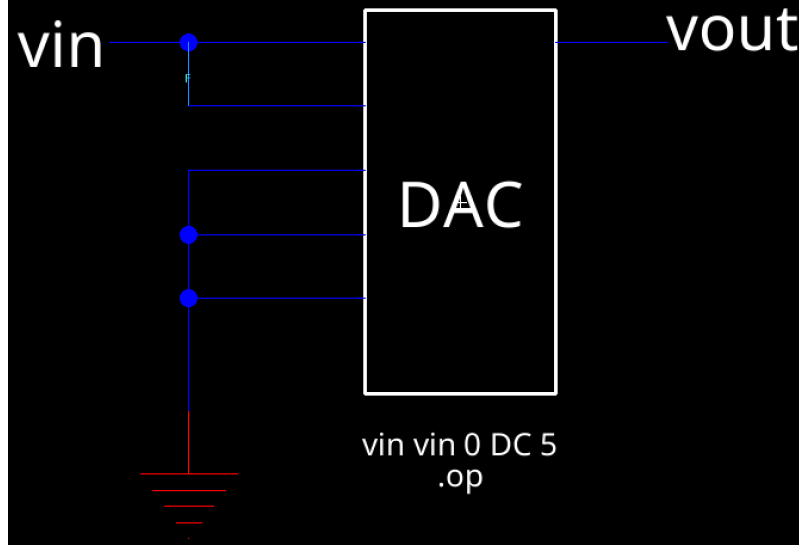


Figure 6: Simplified DAC Schematic Using Electric IC Function

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--- Operating Point ---

V (dac@0_vout) : 0.46875          voltage
V (vin) : 5                      voltage
I (vin) : -0.000248535          device_current
Ix (xdac@0:b0) : 0.000124512      subckt_current
Ix (xdac@0:b1) : 0.000124023      subckt_current
Ix (xdac@0:b2) : -6.44531e-05     subckt_current
Ix (xdac@0:b3) : -3.51562e-05     subckt_current
Ix (xdac@0:b4) : -2.34375e-05     subckt_current
Ix (xdac@0:vout) : 0              subckt_current

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Figure 7: Simplified DAC LTspice Output Results

As seen in the simplified LTspice simulation output, the output result is identical at $0.46875V$; however, the schematic out and output log are much more simplified compared to the full DAC schematic. The output can be mathematically validated by the following:

$$V_{out,b_0,b_1} = (1) \cdot \frac{5}{2^5} + (2) \cdot \frac{5}{2^5} = 0.46875V$$

2.3 Pulse Analysis

The following images detail the pulse response of the circuit when a pulse wave is applied to inputs b_4 and b_3 . Traces placed on the output voltage curve of the LTspice simulation demonstrate that the calculated delay time $t_d = 70ns$ is validated by analysis of the 50% voltage output.

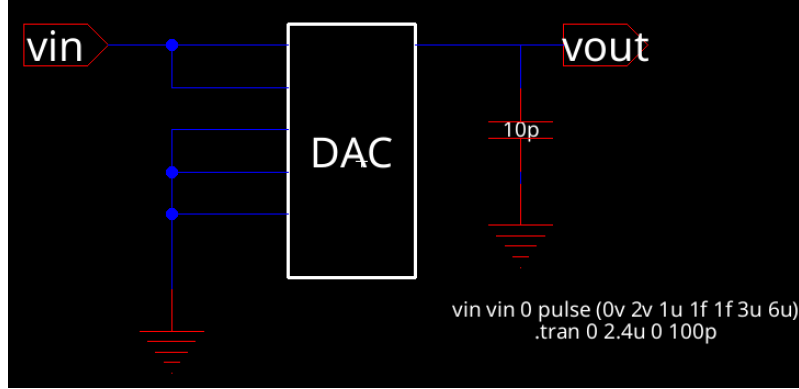


Figure 8: Simplified DAC Schematic with a 10pF connected to V_{out}

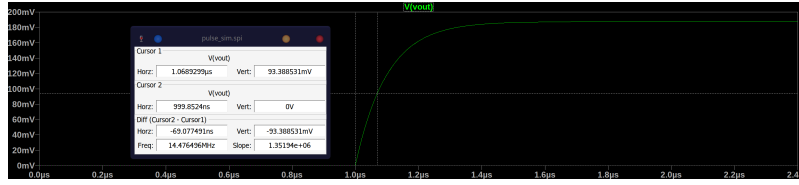


Figure 9: LTspice Pulse Analysis Output with Corresponding Traces Validated Time Delay

2.4 Full DAC

The following figures demonstrate the full R-2R DAC in layout view and 3D view. This layout can be specifically navigated under DAC_simlay.

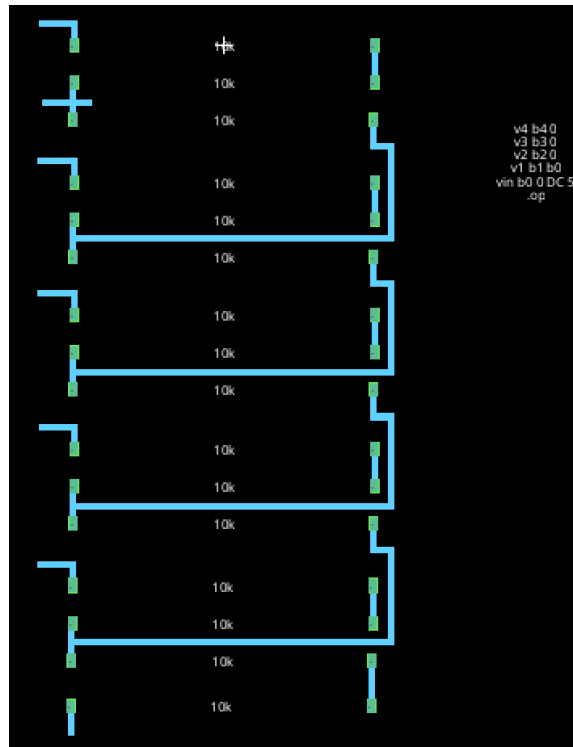


Figure 10: Full DAC in Layout Editor

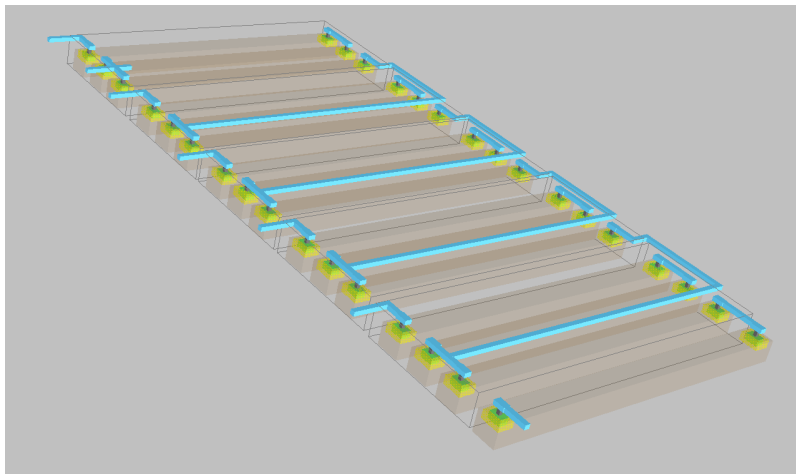


Figure 11: Full DAC in 3D View

3 Documentation

[1] - ENCE_3501_Lab_1.pdf

[2] - Lab_1_DAC.pdf